METRIC
FF-B-171/30
30 NOVEMBER 3, 1993

FEDERAL SPECIFICATION SHEET

BEARING, BALL, ANNULAR, SINGLE ROW, ANGULAR CONTACT, COUNTERBORED OUTER RING, CONTACT ANGLE ABOVE 32° UP TO AND INCLUDING 45°, NON-SEPARABLE, ANSI/AFBMA DIMENSION SERIES 03.

The complete requirements for acquiring the bearings described herein shall consist of this document and the latest issue of FF-B-171/GEN.

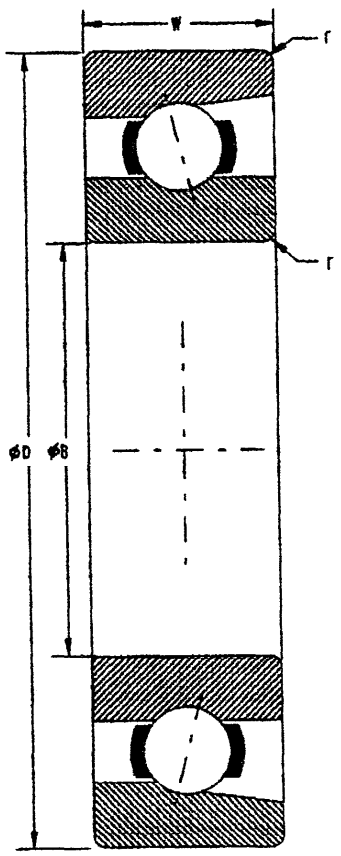


FIGURE 1.
OPEN (NO SHIELDS OR SEALS)
(CODE A)

BEARING CONFIGURATIONS

AMSC/NA FSC 3110

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REQUIREMENTS:

1. **Boundary dimensions.** The boundary dimensions shall be specified in the part number by the size code in table I.

2. Cages. Cages shall be manufactured from any material unless otherwise specified. Cages shall be specified in the part number by the codes in table II.

3. Shields and Seals. All angular contact bearings with counterbored outer rings shall be open (no shields or seals, see figure 1) and designated by the code "A" in the part number.

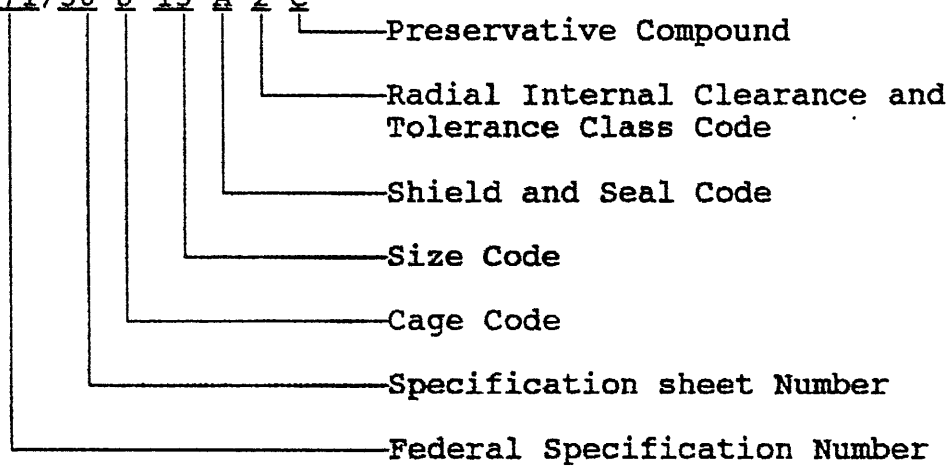
4. Radial Internal Clearance. The tolerance limits for radial internal clearance shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table III.

5. **Tolerance Class.** The tolerance limits for tolerance class ABEC-1 and ABEC-5 shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table III.

6. Preservative Compound. All bearings shall be supplied with preservative compound in accordance with MIL-P-197 and designated by the code "C" in the part number.

7. **Part number.** The part number consists of the letters FFB, the basic Federal specification number, the specification sheet number, and alpha or numeric codes for cage, size, shields, and seals, radial internal clearance and tolerance class, and preservative compound.

EXAMPLE: FFB171/30 J 13 A 2 C



FFB171/30J13A2C indicates - Bearing, cage-pressed steel or molded plastic; bore 65mm; outside diameter 140mm; width 33mm; open (no shields or seals); radial internal clearance symbol 0 (normal); tolerance class ABEC-I; preservative compound in accordance with MIL-P-197.

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TABLE I. Boundary dimensions.

Size Code 1/	OB Bore	OD Outside Diameter	W Width	Chamfer $r_s$min 2/	Shaft Shoulder Diameter Min 3/	C Dynamic Load Rating (Pounds)
00	10	35	11	0.6	15	500
01	12	37	12	1.0	18	1780
02	15	42	13	1.0	21	1985
03	17	47	14	1.0	23	2670
04	20	52	15	1.1	27	2910
05	25	62	17	1.1	32	3770
06	30	72	19	1.1	37	5070
07	35	80	21	1.5	43.5	6190
08	40	90	23	1.5	48.5	7410
09	45	100	25	1.5	53.5	9295
10	50	110	27	2.0	60	10790
11	55	120	29	2.0	65	12385
12	60	130	31	2.1	72	14075
13	65	140	33	2.1	77	15875
14	70	150	35	2.1	82	18305
15	75	160	37	2.1	87	19820
16	80	170	39	2.1	92	21370
17	85	180	41	3.0	99	22950
18	90	190	43	3.0	104	24565
19	95	200	45	3.0	109	26205
20	100	215	47	3.0	114	29505
21	105	225	49	3.0	119	31305
22	110	240	50	3.0	124	33095
24	120	260	55	3.0	134	38890
26	130	280	58	4.0	148	42810
28	140	300	62	4.0	158	46830
30	150	320	76	4.0	168	50960
32	160	340	68	4.0	178	52905
34	170	360	72	4.0	188	57865
36	180	380	75	4.0	198	62410
38	190	400	78	5.0	212	64520
40	200	420	80	5.0	222	70020

1/ Size codes in Table I are industry bore codes.

2/ The minimum bearing chamfer ($r_s$min) must clear the maximum shaft or housing fillet radius. This dimension does not control the shape of the bearing chamfer contour.

3/ For reference purposes only. Maximum shoulder heights shall be such as to permit bearing removal pressure to be applied to the bearing inner ring. In cases where this requirement cannot be met, provisions shall be made for the removal of the bearing from the shaft without damage to the bearing or other parts.

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TABLE II. Codes for Cages.

Code	Cage
J	Pressed steel or molded plastic
M	Machined bronze or brass
T	Machined non-metallic (phenolic)
Y	Pressed brass

TABLE III. Codes for Radial Internal Clearance and ABEC tolerance.

Code	Radial Internal Clearance	Tolerance Class
1	Symbol 2 (Less than normal)	ABEC-1
2	Symbol 0 (Normal)	
3	Symbol 3 (Greater than normal)	
4	Symbol 4 (Greater than symbol 3)	
5	Symbol 2 (Less than normal)	ABEC-5
6	Symbol 0 (Normal)	
7	Symbol 3 (Greater than normal)	
8	Symbol 4 (Greater than symbol 3)	

MILITARY INTERESTS:

Military Coordinating Activity:
DLA - IS

Custodians:
Army - AT
Air Force - 11

Review Activities:
Army - AR, EA, MI
Navy - OS

User Activity:
Army - GL, ME

CIVIL AGENCY COORDINATING ACTIVITIES:

PREPARING ACTIVITY:
DLA-IS

(Project 3110-0848-30)

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